Práctica global

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In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib_venn import venn2
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OrdinalEncoder, OneHotEncoder,
MinMaxScaler, StandardScaler
from sklearn.decomposition import PCA

from sklearn.cluster import KMeans
```

In [2]:
```
coral_ref = pd.read_csv('./coral_reef_sites.csv', sep=',')
ocean_climate = pd.read_csv('./realistic_ocean_climate_dataset.csv',
sep=',')
```

Visualización de información acerca de los datos

In [3]:
```
coral_ref.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 23 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   id                           10000 non-null  int64
 1   region                       10000 non-null  str
 2   depth                        10000 non-null  float64
 3   substrate_type               10000 non-null  str
 4   structural_complexity        10000 non-null  float64
 5   water_temperature            10000 non-null  float64
 6   salinity                     10000 non-null  float64
 7   light_availability           10000 non-null  str
 8   coral_cover                  10000 non-null  float64
 9   algal_cover                  10000 non-null  float64
 10  marine_protection_status     10000 non-null  str
 11  proximity_to_human_activity  10000 non-null  float64
 12  species_Porites              10000 non-null  int64
 13  species_Favia                10000 non-null  int64
 14  species_Stylophora           10000 non-null  int64
 15  species_Acropora             10000 non-null  int64
 16  species_Goniastrea           10000 non-null  int64
 17  species_Montipora            10000 non-null  int64
 18  species_Pocillopora          10000 non-null  int64
 19  species_Montastraea          10000 non-null  int64
 20  species_Orbicella            10000 non-null  int64
 21  species_Siderastrea          10000 non-null  int64
 22  species_Fungia               10000 non-null  int64
dtypes: float64(7), int64(12), str(4)
memory usage: 1.8 MB
```

In [4]:
```
ocean_climate.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Date                500 non-null    str
 1   Location            500 non-null    str
 2   Latitude            500 non-null    float64
 3   Longitude           500 non-null    float64
 4   SST (°C)            500 non-null    float64
 5   pH Level            500 non-null    float64
 6   Bleaching Severity  350 non-null    str
 7   Species Observed    500 non-null    int64
 8   Marine Heatwave     500 non-null    bool
dtypes: bool(1), float64(4), int64(1), str(3)
memory usage: 31.9 KB
```

Unificación de las regiones

In [5]:
```
print(set(ocean_climate['Location'].unique()))
print(set(coral_ref['region'].unique()))
```

```
{'Red Sea', 'Caribbean Sea', 'Galápagos', 'Maldives', 'Hawaiian
Islands', 'Great Barrier Reef', 'South China Sea'}
{'Red Sea', 'Caribbean', 'Great Barrier Reef', 'Coral Triangle'}
```

In [6]:
```
ocean_climate['Location'] = ocean_climate['Location'].apply(
    lambda x: x.lower().replace(' ', '_')
)

coral_ref['region'] = coral_ref['region'].apply(
    lambda x: x.lower().replace(' ', '_') + ('_sea' if x == 'Caribbean'
else '')
)

set(ocean_climate['Location'].unique()).union(set(coral_ref['region'].un
ique()))
```

Out[6]:
```
{'caribbean_sea',
 'coral_triangle',
 'galápagos',
 'great_barrier_reef',
 'hawaiian_islands',
 'maldives',
 'red_sea',
 'south_china_sea'}
```

Eliminación de columnas no necesarias

In [7]:
```
ocean_climate_reduced = ocean_climate.drop(columns = ['Date'])

coral_ref_reduced = coral_ref.drop(columns = [
        'id', 'depth', 'substrate_type', 'structural_complexity',
        'coral_cover', 'algal_cover', 'marine_protection_status',
    'proximity_to_human_activity'
])
```

Pregunta 1

- Principales diferencias en las variables meteorológicas entre el mar Rojo y el Caribe
- `SST` , `water_temperature` , `light_availability` , `salinity`

- Como se refleja las diferencias en las especies (`coral_ref`)
- `species_*`

In [8]:

```
# Agrupación de variables de los datos para la respuestas
columnas_especies = [
    columna for columna in coral_ref_reduced.columns if
columna.startswith('species')
]

coral_ref_apt_1 = coral_ref_reduced[
    ['region', 'water_temperature', 'light_availability'] +
columnas_especies
].loc[
    (coral_ref_reduced['region'] == 'red_sea')
    | (coral_ref_reduced['region'] == 'caribbean_sea')
]

ocean_climate_apt_1 = ocean_climate_reduced[
    ['Location', 'SST (°C)', 'Marine Heatwave']
].loc[
    (ocean_climate_reduced['Location'] == 'red_sea')
    | (ocean_climate_reduced['Location'] == 'caribbean_sea')
]

##
################################################################################
#####
## Parte 1 de la pregunta

agr_coral_ref = (
    coral_ref_apt_1[['region', 'water_temperature',
'light_availability']]
).groupby('region')

light_availability = (coral_ref_apt_1[['region',
'light_availability']]).value_counts(sort=False)

agr_ocean_climate = (
    ocean_climate_apt_1[['Location', 'SST (°C)']]
).groupby('Location')

marine_heatwave = (ocean_climate_apt_1[['Location', 'Marine
Heatwave']]).value_counts(sort=False)

##
################################################################################
#####
## Parte 2 de la pregunta

# El campo de las especies significa que se ha encontrado un individuo
de dicha
# especie. De esta forma, si se agrupa por región y se obtiene el
máximo, podemos
# obtener si se ha encontrado al menos un individuo de dicha especia en
esa región
especies_mares = (coral_ref_apt_1[['region'] +
columnas_especies]).groupby('region').max()

especies = np.array(columnas_especies)
```

```
especies_red_sea = {
    especie.split("_")[1] for especie in
especies[(especies_mares.loc['red_sea'] == 1).values]
}

especies_caribbean_sea = {
        especie.split("_")[1] for especie in
especies[(especies_mares.loc['caribbean_sea'] == 1).values]
}
```

Resultados intermedios

Valores que serán representados en las gráficas

In [9]:
```
# Temperatura del agua
agr_coral_ref['water_temperature'].describe()
```

Out[9]:

	count	mean	std	min	25%	50%	75
region							
caribbean_sea	2498.0	26.972578	1.169495	25.0	26.0	27.0	28
red_sea	2573.0	27.952274	1.141092	26.0	26.9	28.0	28

In [198]:
```
# Temperatura en la superficie del agua
agr_ocean_climate['SST (°C)'].describe()
```

Out[198]:

	count	mean	std	min	25%	50%
Location						
caribbean_sea	70.0	28.681143	1.509358	24.86	27.88	28.86
red_sea	83.0	28.394458	1.453663	24.86	27.53	28.31

In [11]:
```
# Disponibilidad de luz en los momentos en los que se hace la medición
light_availability
```

Out[11]:
```
region         light_availability
red_sea        Low                   1038
               Medium                1051
               High                   484
caribbean_sea  High                   799
               Low                    544
               Medium                1155
Name: count, dtype: int64
```

In [12]:
```
# Distribución de olas de calor en los días en los que se hace la
medición
marine_heatwave.sort_index()
```

Out[12]:
```
Location       Marine Heatwave
caribbean_sea  False             57
               True              13
red_sea        False             72
               True              11
Name: count, dtype: int64
```

In [13]:
```
especies_red_sea
```

Out[13]:
```
{'Acropora', 'Favia', 'Porites', 'Stylophora'}
```

In [14]:
```
especies_caribbean_sea
```

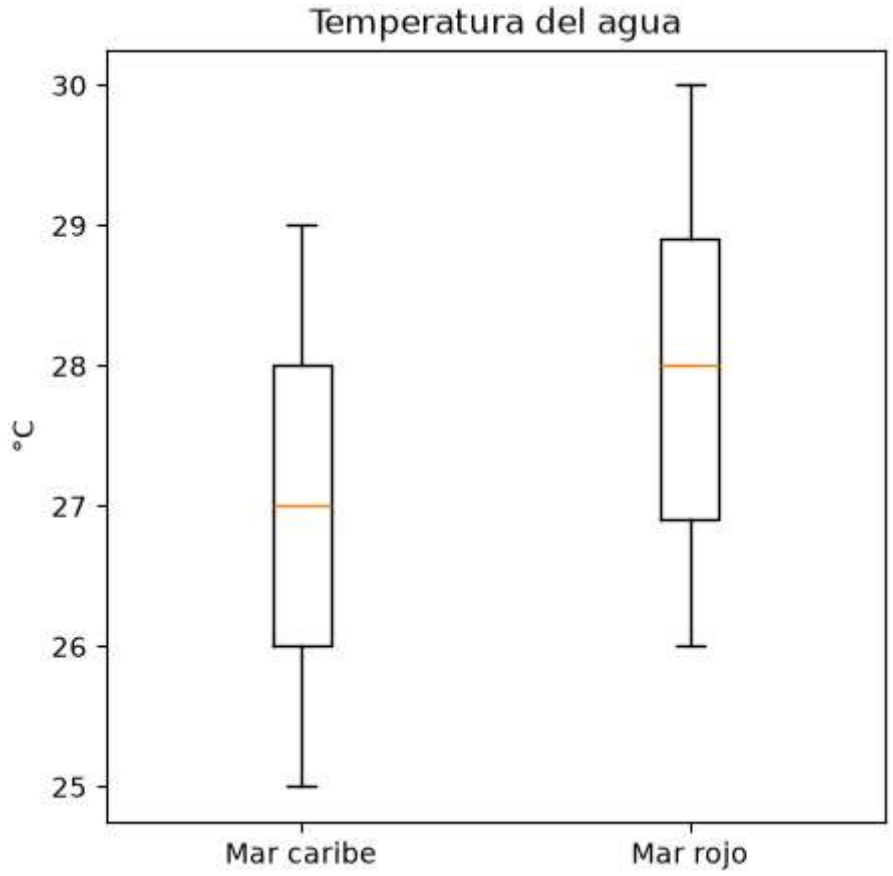
Out[14]:
```
{'Montastraea', 'Orbicella', 'Porites', 'Siderastrea'}
```

Visualización de estos datos

In [15]:
```
fig, ax = plt.subplots(1,1, figsize=(5,5))

ax.boxplot(
    x           = [grupo.values for nombre, grupo in
agr_coral_ref['water_temperature']],
    tick_labels = ['Mar caribe', 'Mar rojo']
)
ax.set_title('Temperatura del agua')
ax.set_ylabel('°C')

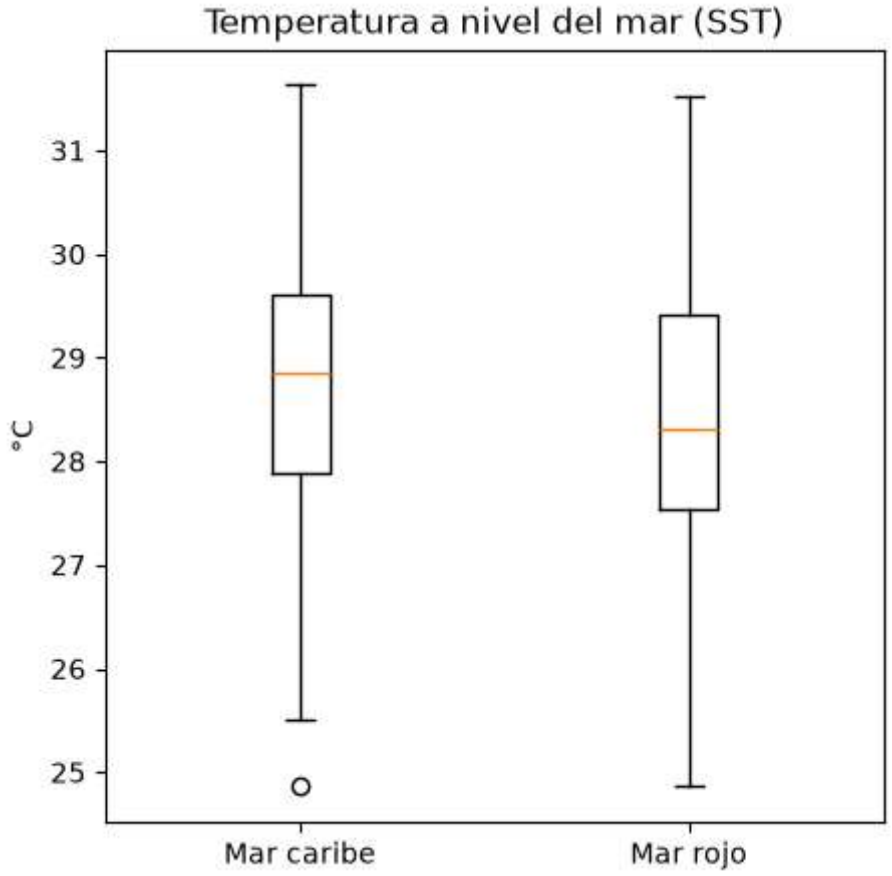
plt.show()
```



In [16]:
```
fig, ax = plt.subplots(1,1, figsize=(5,5))

ax.boxplot(
    x = [grupo.values for nombre, grupo in agr_ocean_climate['SST
(°C)']],
    tick_labels=['Mar caribe', 'Mar rojo']
)
ax.set_title('Temperatura a nivel del mar (SST)')
ax.set_ylabel('°C')

plt.show()
```



In [17]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 5))

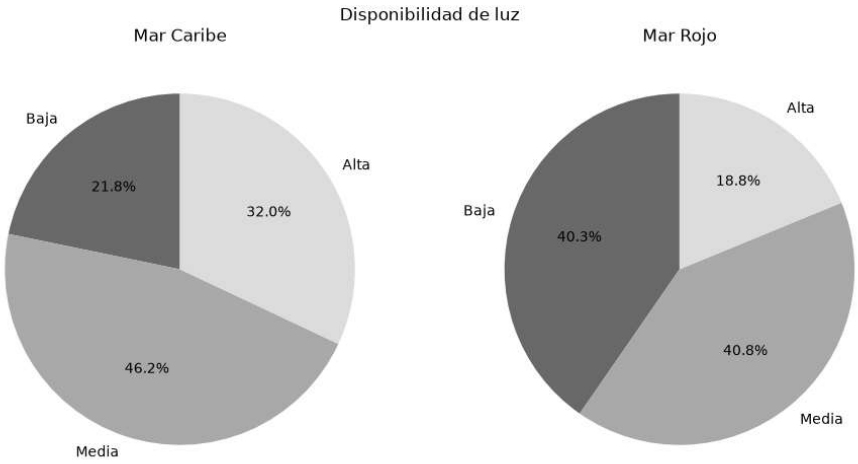
etiquetas = ['Baja', 'Media', 'Alta']

# Diagrama de tarta: Mar Caribe
ax1.pie(
    light_availability.caribbean_sea[['Low', 'Medium', 'High']],
    labels=etiquetas,
    autopct='%1.1f%%',
    startangle=90,
    colors=['dimgray', 'darkgrey', 'gainsboro']
)
ax1.set_title('Mar Caribe')

# Diagrama de tarta: Mar Rojo
ax2.pie(
    light_availability.red_sea[['Low', 'Medium', 'High']],
    labels=etiquetas,
    autopct='%1.1f%%',
    startangle=90,
    colors=['dimgray', 'darkgrey', 'gainsboro']
)
ax2.set_title('Mar Rojo')

fig.suptitle('Disponibilidad de luz')

plt.tight_layout()
plt.show()
```



In [18]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 5))

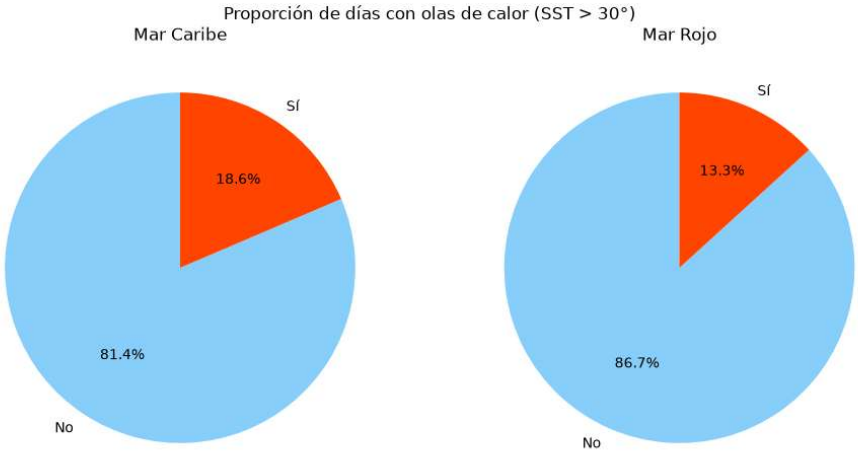
etiquetas = ['No', 'Sí']

ax1.pie(
    marine_heatwave.caribbean_sea,
    labels=etiquetas,
    autopct='%1.1f%%', # Muestra el porcentaje con 1 decimal
    startangle=90,     # Gira el gráfico para que empiece en vertical
    colors=['lightskyblue', 'orangered']
)
ax1.set_title('Mar Caribe')

ax2.pie(
    marine_heatwave.red_sea,
    labels=etiquetas,
    autopct='%1.1f%%',
    startangle=90,
    colors=['lightskyblue', 'orangered']
)
ax2.set_title('Mar Rojo')

fig.suptitle('Proporción de días con olas de calor (SST > 30°)')

plt.tight_layout()
plt.show()
```



In [ ]:
```
plt.figure(figsize=(6, 6))

diagrama = venn2(
    [especies_red_sea, especies_caribbean_sea],
    ('Mar rojo', 'Mar caribe'),
    set_colors = ('red', 'deepskyblue')
)

# Caso del mar rojo
diagrama.get_label_by_id('10').set_text('\n\n'.join(especies_red_sea -
especies_caribbean_sea))
# Caso del mar caribe
diagrama.get_label_by_id('01').set_text('\n\n'.join(especies_caribbean_s
ea - especies_red_sea))
# Intersección
diagrama.get_label_by_id('11').set_text('\n'.join(especies_caribbean_sea
& especies_red_sea))

plt.title("Especies por mar")

plt.show()
```



Pregunta 2

- Comparación de regiones entre SST y número de especies para buscar patrón
- Número de especies vs variación del SST media

- Comparación de regiones entre SST y especies por sitio para buscar patrón
- valor del SST y especies existentes

In [62]:
```
columnas_especies = [
    columna for columna in coral_ref_reduced.columns if
columna.startswith('species')
]

# Datos usados para la respuesta a la pregunta
ocean_climate_apt_2 = ocean_climate_reduced[
    ['Location', 'SST (°C)', 'Marine Heatwave']
]

coral_ref_apt_2 = coral_ref_reduced[['region'] + columnas_especies]

##
#############################################################################
#####
## Parte 1 de la pregunta

medias_climatica = ocean_climate_apt_2.groupby('Location').agg(
    mean_sst = ('SST (°C)', 'mean'),
    porc_ola_calor = ('Marine Heatwave', 'mean'),
    total_olas_calor = ('Marine Heatwave', 'sum')
).reset_index().set_index('Location')

existencia_especies = coral_ref_apt_2.groupby('region').max()

numero_especies = coral_ref_apt_2.groupby('region').max().sum(axis=1)
numero_especies.name = 'num_especies'

info_mares =
medias_climatica.join(numero_especies).join(existencia_especies).dropna(
axis='index')
```

Resultados intermedios

Valores que serán representados en las gráficas

In [63]:
```
info_mares
```

Out[63]:

	mean_sst	porc_ola_calor	total_olas_calor	num_es
Location				
caribbean_sea	28.681143	0.185714	13	4.0
great_barrier_reef	28.469195	0.091954	8	4.0
red_sea	28.394458	0.132530	11	4.0

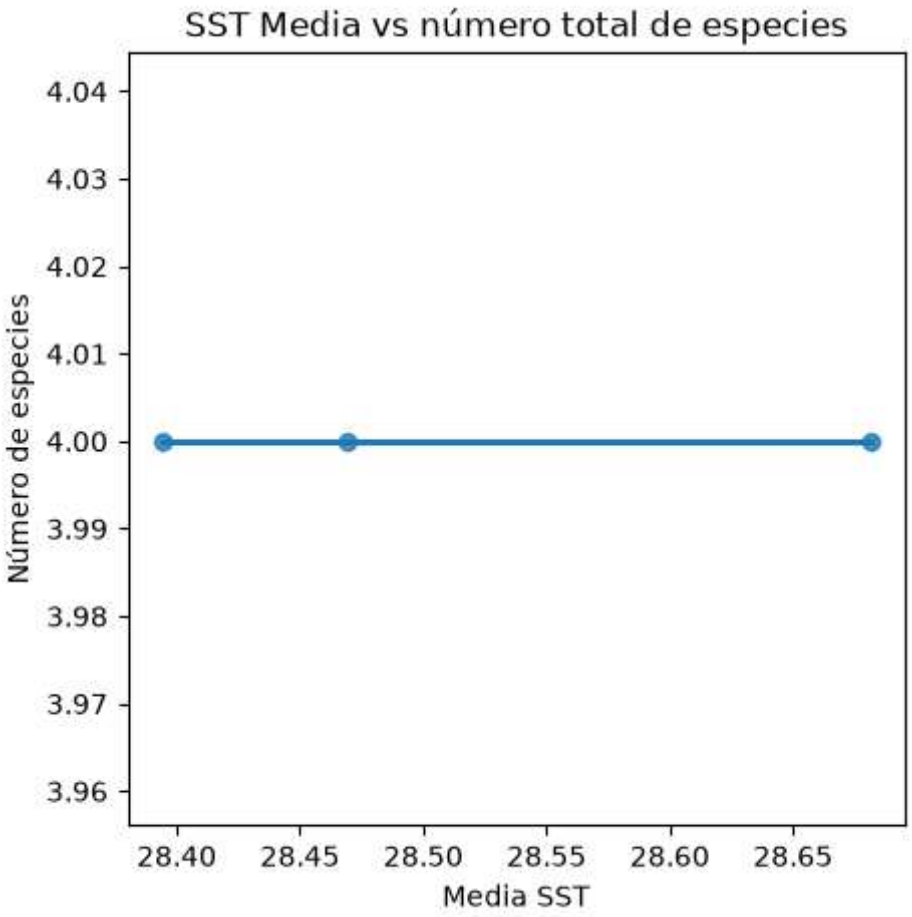
Visualización de estos datos

In [101]:
```
fig, ax = plt.subplots(1, 1, figsize=(5, 5))

sns.regplot(data=info_mares, x='mean_sst', y='num_especies', ax=ax)
ax.set_title('SST Media vs número total de especies')
ax.set_xlabel('Media SST')
ax.set_ylabel('Número de especies')

plt.setp(ax.collections[1], alpha=0)

plt.show()
```

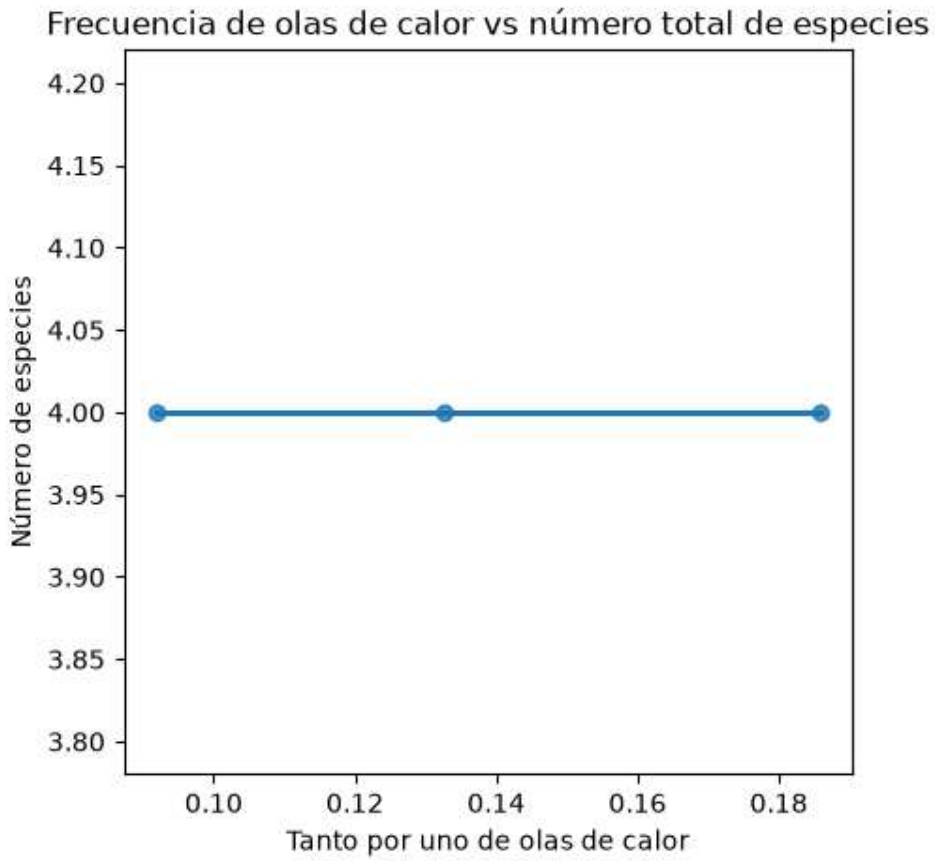


In [102]:

```
fig, ax = plt.subplots(1, 1, figsize=(5, 5))

sns.regplot(data=info_mares, x='porc_ola_calor', y='num_especies',
ax=ax, ci=0)
ax.set_title('Frecuencia de olas de calor vs número total de especies')
ax.set_xlabel('Tanto por uno de olas de calor')
ax.set_ylabel('Número de especies')

plt.show()
```



In [85]:
```
corr_matrix = info_mares[['mean_sst', 'porc_ola_calor'] +
columnas_especies].corr()
corr_composition = corr_matrix.loc[['mean_sst', 'porc_ola_calor'],
columnas_especies]

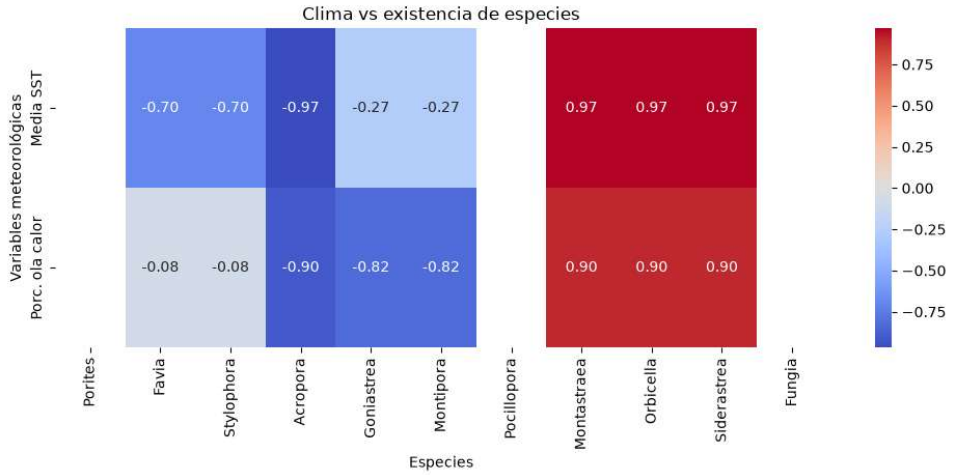
fig, ax = plt.subplots(1,1, figsize=(12, 4))

sns.heatmap(corr_composition, annot=True, cmap='coolwarm', center=0,
fmt=".2f")

ax.set_title('Clima vs existencia de especies')
ax.set_ylabel('Variables meteorológicas')
ax.set_xlabel('Especies')

ax.set_xticklabels([columna.split("_")[1] for columna in
columnas_especies])
ax.set_yticklabels(['Media SST', 'Porc. ola calor'])

plt.show()
```



Pregunta 3

- Relación entre pH y blanqueaminto de los corales
- ¿Preprocesado de los datos para completar?

In [ ]:
```
ocean_climate_apt_3 = ocean_climate[['Location', 'pH Level', 'Bleaching
Severity']]

ocean_climate_apt_3 = ocean_climate_apt_3.dropna(axis=0)

ocean_climate_apt_3['Bleaching Severity'] = pd.Categorical(
    ocean_climate_apt_3['Bleaching Severity'],
    categories=['Low', 'Medium', 'High'],
    ordered=True
)
```

Resultado intermedio

In [189]:
```
ocean_climate_apt_3.groupby(['Location', 'Bleaching Severity']).mean()
```

Out[189]:

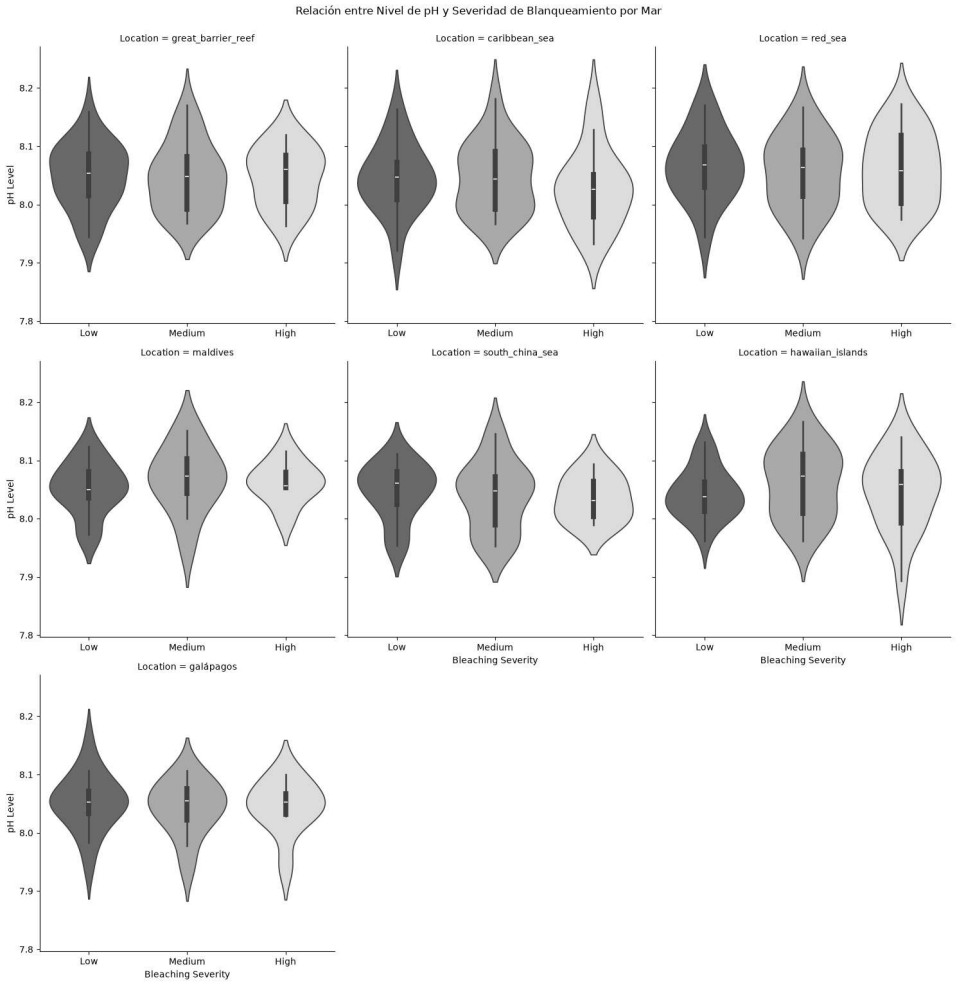
		pH Level
Location	Bleaching Severity	
caribbean_sea	Low	8.046000
	Medium	8.046789
	High	8.027800
galápagos	Low	8.052421
	Medium	8.042846
	High	8.045333
great_barrier_reef	Low	8.048625
	Medium	8.047083
	High	8.047786
hawaiian_islands	Low	8.043500
	Medium	8.062118
	High	8.045687
maldives	Low	8.052684
	Medium	8.068583
	High	8.062857
red_sea	Low	8.066267
	Medium	8.055611
	High	8.062316
south_china_sea	Low	8.047053
	Medium	8.038556
	High	8.034250

Visualización

In [187]:
```
sns.catplot(
    data        = ocean_climate_apt_3,
    x           = 'Bleaching Severity',
    hue         = 'Bleaching Severity',
    y           = 'pH Level',
        col             = 'Location',
    palette     = ['gainsboro', 'darkgrey', 'dimgray'],
    col_wrap = 3,
    sharex      = False,
    kind        = "violin",
    dodge       = False
)

plt.suptitle('Relación entre Nivel de pH y Severidad de Blanqueamiento
por Mar', y=1.02)

plt.show()
```



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